
Mechanistic Analysis of Future Representations in World Action Models

Nicole Ma

Abstract

World action models predict both how a scene will change and what action a robot should take. We study whether their predicted futures actually influence their actions. Previous work found that corrupting cache values assigned to future positions changes model behavior, but corruption cannot show what information the model uses. To test the role of future content directly, we obtain two possible continuations from the same simulator state. We then recompute one action after substituting the future representation from the other continuation, while holding the observation, instruction, noise, solver, and denoising schedule fixed. We measure how far the recomputed action moves from its original value toward the action paired with the replacement future; 0 indicates no movement and 1 reproduces the alternative action. We use the same normalization for simulator endpoints. At the first Cosmos Policy state, the recomputed action moves 0.499 of the recipient-to-donor displacement and the executed endpoint moves 0.552, with positive effects in all ten LIBERO-10 tasks. These effects are much smaller at later states. Additional Cosmos Policy experiments find positive action effects when wrist- or primary-camera futures are replaced, but near-zero effects when future proprioception or object state alone is changed. In Cosmos 3, replacement futures also move actions and simulator endpoints toward the corresponding alternative across four RoboLab tasks. Across four calibration tasks, replacing the transformer-attention keys and values associated with the replacement future at every direct future-to-action connection reduces this action movement by 0.842 projection units. Thus, changing only the predicted future changes action in a way that depends on the future’s content. The Cosmos Policy experiments are most consistent with the model using predicted visible robot motion, while the separate Cosmos 3 experiment shows that future-token keys and values contribute to the action change. Neither experiment shows that the models compare predicted task outcomes when choosing an action.

1 Introduction

World action models (WAMs) generate predicted future observations and robot actions within the same model [8, 14]. We study two such models. Cosmos Policy represents actions and future observations in a common latent sequence, while Cosmos 3 jointly denoises future-video and action tokens [8, 14]. In both cases, the action computation can use information from the predicted future. The model architecture does not tell us whether it does so, or what information it uses when it does.

Changing the representation of a future could affect action for several reasons. Future tokens may participate in the model’s computation without describing a particular behavior; disrupting them could then change the action in an arbitrary direction. The model might instead use predicted video of the robot’s motion to help generate the corresponding action. A more planning-like possibility is that the model predicts changes in task objects and uses those consequences to choose among actions. Explicit visual planners implement the latter approach by generating candidate actions or subgoals, predicting their outcomes, and comparing those outcomes using a goal or value function

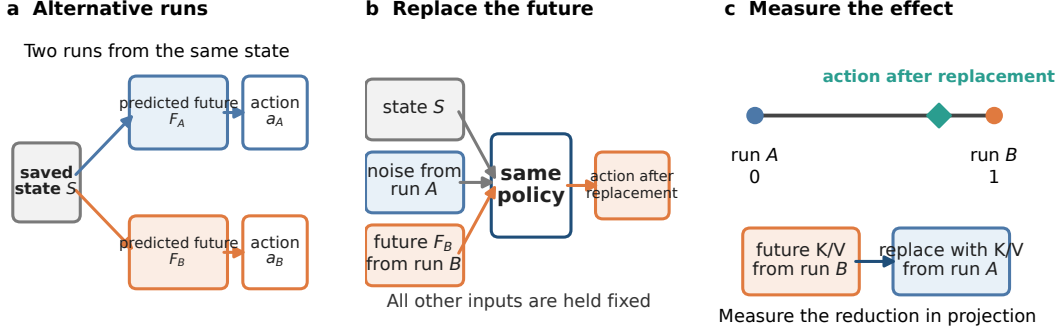


Figure 1: Future transplantation. (a) The policy generates two continuations from the same simulator state. (b) We recompute the recipient with the donor’s predicted future while keeping its current inputs and noise fixed. (c) We measure how far the resulting action and endpoint move from the recipient toward the donor. In the Cosmos 3 pathway experiment, we then replace the keys and values associated with future tokens to test whether they contribute to this change.

[2, 3, 8]. Cosmos Policy and Cosmos 3 do not use this type of explicit candidate search during the inference procedures studied here. Whether their predicted futures nevertheless provide information about actions or task consequences must therefore be tested experimentally.

RIFT provides the closest prior evidence that future representations affect WAM behavior [21]. It masks cached values stored at future positions or reassigns them to different positions. Across four models evaluated on the 40-task LIBERO Spatial, Object, Goal, and Long protocol, these interventions change closed-loop trajectories and reduce task success. This shows that values associated with future positions, and their locations, matter for action generation. It does not show what information in those values matters. Masking a representation can change an action in any direction, so the resulting change cannot be associated with the meaning of a specific alternative future.

We instead compare two futures generated by the same policy from the same simulator state. Starting from an exactly restored simulator state, we run the unmodified policy twice to obtain two continuations. We call the continuation that will be recomputed the recipient and the other continuation the donor. Each continuation contains a predicted future, an action, and an endpoint obtained by executing that action in the simulator. We recompute the recipient after replacing its predicted future with the donor future, while keeping the current observation, instruction, noise, solver, and denoising schedule unchanged. The donor action and endpoint are never given to the model; they are used only to evaluate the result. We refer to this procedure as reachable-donor future transplantation, or future transplantation for short.

We measure whether the recomputed action and endpoint move from the original recipient behavior toward the donor behavior. An unchanged recipient has a normalized projection of 0, and a result equal to the donor has a projection of 1. A positive difference between the donor-future and self-future conditions therefore means that the replacement moves behavior in the direction associated with the replacement future. We also reverse the roles of the two continuations and, in Cosmos 3, compare several donors generated from the same state. These comparisons test whether the action follows the selected future rather than merely moving away from its original value.

In both models, replacing a future shifts actions and simulator endpoints toward the behavior that originally accompanied the replacement. In Cosmos Policy, the effect is largest at the first evaluation state and is much smaller at later states. We then repeat the intervention separately for future wrist video, primary-camera video, and proprioception. We also change robot and object content separately. The measured effects are most consistent with the model using predicted visible robot motion at the evaluated states; changing object state alone has almost no effect. In a separate Cosmos 3 experiment, replacing the donor run’s future-token keys and values with those from the recipient run substantially reduces the action change. This result identifies an internal representation that contributes to the Cosmos 3 effect, but it does not determine what those tensors represent. The experiments do not test candidate search or value comparison, do not establish consistent effects on task success, and do not show that object consequences are unused in other models or states.

78 2 Background and Related Work

79 2.1 Future prediction in robot policies

80 Robot policies use predicted visual futures in two common ways. Some methods first predict a future
81 video and then infer actions from that video or its representation [1, 7]. Other methods represent
82 video and actions in a shared generative model [6, 8, 10, 14]. Joint training does not necessarily
83 mean that the model must generate or use a future at inference time. UVA, for example, can omit
84 video generation during inference even though it is trained jointly on video and actions [10].

85 Several recent studies separate predictive training from the use of predictions during action gener-
86 ation. Fast-WAM removes explicit future rollout while retaining competitive policy performance
87 [20]. Faster-WAM keeps a lower-cost connection from future representations to actions and reports
88 improved robustness in its evaluation settings [23]. AGRA finds that visually plausible predictions
89 do not necessarily support reliable action inference in one WAM architecture [17]. RIFT directly
90 changes representations associated with future positions and shows that they affect policy behavior
91 [21]. Our experiments ask a different question: whether replacing one policy-generated future with
92 another changes the action toward the behavior associated with the replacement.

93 We first test this question using complete futures. We then repeat the experiment after changing only
94 the predicted video view, proprioception, robot motion, or object state in Cosmos Policy. Finally,
95 we replace internal keys and values associated with future-video tokens in Cosmos 3 to test whether
96 those tensors contribute to the action change.

97 2.2 Activation replacement

98 Information can be present in an activation without affecting the model’s output. Activation-
99 replacement methods test for an effect by substituting an internal representation recorded in one
100 run for the corresponding representation in another run [5, 12, 16, 22]. The rest of the second com-
101 putation is left unchanged. Replacing a group of activations can affect several interacting computa-
102 tional routes, and unmodified representations may compensate for or interact with the replacement
103 [18]. We therefore use the Cosmos 3 experiment only to test whether future-token keys and val-
104 ues contribute to the action change. It does not provide a complete decomposition of the action
105 computation.

106 3 Methods

107 We test whether changing a model’s predicted future changes its action in the direction associated
108 with that future. Starting from the same saved simulator state, we sample two continuations from the
109 unmodified policy. We then rerun one continuation after replacing its future representation with the
110 future from the other continuation. The second continuation provides both the replacement future
111 and an independently generated action against which we evaluate the change.

112 3.1 Future transplantation

113 Let S denote a saved simulator state. Two runs of the unmodified policy from S produce branches
114 A and B , with future representations F_A and F_B , action chunks $\mathbf{a}_A$ and $\mathbf{a}_B$, and executed endpoints
115 $\mathbf{e}_A$ and $\mathbf{e}_B$. We refer to A as the recipient branch and B as the donor branch.

116 To recompute branch A , we restore the same simulator state and current observation. We keep the
117 instruction, action-noise realization, future-noise realization, solver, and denoising schedule fixed,
118 but replace F_A with F_B . The model never receives $\mathbf{a}_B$ or $\mathbf{e}_B$; these values are used only after
119 inference to evaluate the result. We also repeat the experiment with B as the recipient and A as the
120 donor.

121 3.2 Measuring movement toward the donor behavior

122 For an intervened action chunk $\hat{\mathbf{a}}$, we measure its projection onto the direction from the original
123 recipient action to the original donor action:

$$\pi_a(\hat{\mathbf{a}}; A, B) = \frac{(\hat{\mathbf{a}} - \mathbf{a}_A)^\top (\mathbf{a}_B - \mathbf{a}_A)}{\|\mathbf{a}_B - \mathbf{a}_A\|_2^2}. \quad (1)$$

124 Under this normalization, the recipient action has projection 0 and the donor action has projection 1.
125 We define the corresponding projection for an executed endpoint $\hat{\mathbf{e}}$ as

$$\pi_e(\hat{\mathbf{e}}; A, B) = \frac{(\hat{\mathbf{e}} - \mathbf{e}_A)^\top (\mathbf{e}_B - \mathbf{e}_A)}{\|\mathbf{e}_B - \mathbf{e}_A\|_2^2}. \quad (2)$$

126 We execute every intervened action from the same saved simulator state. As a control, we also
127 recompute the recipient with its own future. We report the difference between the donor-future and
128 self-future conditions:

$$\Delta_a = \pi_a(\hat{\mathbf{a}}_{\text{donor}}; A, B) - \pi_a(\hat{\mathbf{a}}_{\text{self}}; A, B), \quad \Delta_e = \pi_e(\hat{\mathbf{e}}_{\text{donor}}; A, B) - \pi_e(\hat{\mathbf{e}}_{\text{self}}; A, B). \quad (3)$$

129 A positive value means that the donor future moves the output from branch A toward branch B .
130 An arbitrary perturbation can instead produce zero or negative projection even when it changes the
131 action substantially.

132 3.3 Models and evaluation sets

133 We use this projection measure for both public policies. For Cosmos Policy, we evaluate the
134 Cosmos-Policy-LIBERO-Predict2-2B checkpoint on the ten long-horizon LIBERO tasks
135 known as LIBERO-10 [8, 11]. The evaluation set was specified before the intervention results were
136 analyzed and contains 30 simulator states: one state from each task at the first policy query, after
137 three action chunks, and after six action chunks. At each state, eight unmodified policy runs are
138 used to select one pair of branches. We evaluate both transplant directions with three future-noise
139 draws each.

140 For Cosmos 3, we evaluate the 16B Cosmos3-Nano-Policy-DROID checkpoint in RoboLab
141 [14, 15, 19]. We select one simulator state from each of four tasks without using intervention out-
142 comes. The four states provide 12 donor continuations in total. We test each donor using two
143 versions of its future: the future predicted by the model and a future obtained by encoding the video
144 produced when the donor action is executed in the simulator. We reserve a fifth state from the
145 Banana task for the held-out experiment in Section 6.

146 3.4 Controls and statistical analysis

147 We select donor pairs using only the distance between their unmodified actions and endpoints. In
148 addition to the self-future control, we compare against Gaussian noise matched to the donor future
149 in norm and distance from the recipient future, a future shuffled within the same task, and a natural
150 control chosen before evaluation. The available experiment record does not specify how the natural
151 control was constructed. No-op recomputations verify that the intervention code leaves an unmod-
152 ified run unchanged. Our unit of statistical independence is an exactly restored simulator state.
153 Reciprocal directions, donors, noise draws, and repeated process launches are treated as repeated
154 measurements of that state.

155 We first average the measurements within each state and then compute 95% confidence intervals
156 using 10,000 cluster-bootstrap resamples over states [4]. For the Cosmos 3 experiment, which con-
157 tains one independent state per task, the bootstrap resamples the four tasks. For the confirmatory
158 Cosmos Policy Predict2 experiments, the prespecified smallest effect of interest is 0.10 projection
159 units, or ten percent of the displacement from recipient to donor [9]. Appendix A provides additional
160 implementation and reproducibility details.

4 Results

4.1 Cosmos Policy

At the first policy query, the mean difference in action projection between the donor-future and self-future conditions is 0.499 (95% CI [0.336, 0.679]) and is positive in all ten tasks. The corresponding difference in executed endpoint projection is 0.552 ([0.387, 0.728]), also positive in all ten tasks.

We also compare the donor-future condition with three other controls. The action-projection differences relative to the Gaussian, natural, and shuffled-future controls are 0.497, 0.103, and 0.255, respectively. The corresponding endpoint-projection differences are 0.549, 0.110, and 0.277. Confidence intervals are not available for these control comparisons. Thus, at the first policy query, replacing the original future with a donor future moves both the action and the executed endpoint toward the donor continuation.

The effect is much smaller at the two later evaluation states (Figure 2a). After three action chunks, the mean differences are 0.031 for action projection and 0.004 for endpoint projection. After six chunks, they are 0.027 and 0.025. Across all 30 states, the mean action effect is 0.186 ([0.095, 0.290]) and the mean endpoint effect is 0.194 ([0.091, 0.305]). The lower bounds of both intervals fall just below the prespecified threshold of 0.10. Each evaluation point uses a different saved state, so the comparison does not separate the effect of elapsed time from other differences among the states. We therefore observe a large effect at the sampled first-query states, but cannot conclude that the effect weakens because time has elapsed.

4.2 Cosmos 3

Cosmos3-Nano has 36 transformer layers and jointly denoises future-video and action tokens using bidirectional diffusion attention [14, 15]. At the final boundary of the guided sampler, we replace only the unconditioned future-video coordinates; no action coordinate is modified. Control runs reproduce the original action bitwise and confirm that the intervention never writes directly to the action tokens. To evaluate endpoints, we reconstruct the RoboLab simulator, restore the saved state, and replay the same observation prefix before executing each action [19].

Across the four tasks, transplanting model-predicted donor futures produces a mean action projection of 0.985 (task-bootstrap 95% CI [0.974, 0.993]) and a mean simulator-endpoint projection of 0.975 ([0.833, 1.079]). Transplanting future representations encoded from simulator-executed donor branches produces a mean action projection of 1.060 ([1.010, 1.103]) and a mean endpoint projection of 1.027 ([0.824, 1.204]). All 24 interventions, comprising 12 donors evaluated with two future sources, have positive projection. For each intervention, we also ask which native donor is closest to the resulting output. Action-space nearest-donor accuracy is 12/12 for both future sources. Endpoint nearest-donor accuracy is 10/12 for predicted futures and 11/12 for futures encoded from executed branches. The self-future and geometry-matched Gaussian controls have projections near zero, although exact confidence intervals are not available for these controls.

The Cosmos 3 results are based on four independent simulator states, one each from the Rubik’s cube, mustard, spoon, and marker tasks. Testing several donors from each state determines whether the effect is specific to the chosen donor, but does not increase the number of independent states. We use a fifth state from the Banana task only for the held-out test in Section 6. None of the selected branch pairs in either model showed a stable difference in binary task success across repeated continuations. We therefore report effects on actions and endpoints, not on task success.

5 Analysis of Future Content in Cosmos Policy

A donor future changes several correlated parts of the predicted scene, including robot pose, object state, camera motion, and task progress. We conduct three additional experiments in Cosmos Policy to determine which changes are associated with the action effect.

5.1 Visual and proprioceptive futures

We first repeat the intervention while replacing one component of the future at a time. At the ten first-query states, replacing the wrist-camera future produces a mean action effect of 0.435

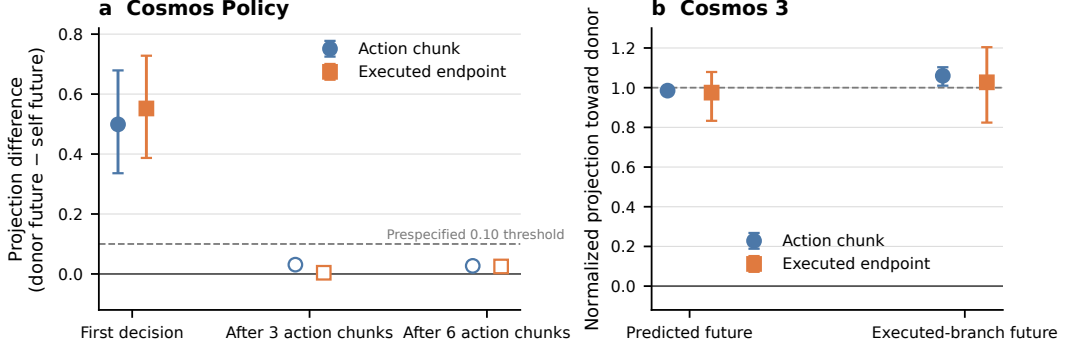


Figure 2: Results of future transplantation. (a) In Cosmos Policy, action and endpoint projection are largest at the first policy query and much smaller at the two later evaluation states. Intervals are shown where available. (b) All 24 Cosmos 3 interventions have positive projection toward their respective donors. Bars show 95% confidence intervals obtained by resampling tasks.

([0.284, 0.605]). Replacing the primary-camera future produces a smaller mean effect of 0.101 ([0.057, 0.144]). Replacing future proprioception produces a mean effect of 0.001 ([−0.004, 0.008]). Thus, the largest effect occurs for the wrist camera, which closely tracks the manipulator, while the confidence interval for future proprioception includes zero (Figure 3a).

5.2 Separating robot motion from object state

To separate robot motion from object state, we analyze a 2×2 experiment that was specified before analysis. At 20 held-out LIBERO states, robot state $R \in \{0, 1\}$ and object state $O \in \{0, 1\}$ are re-rendered independently, producing all four combinations. A decoder recovers the intended combination with 92.5% top-1 accuracy from the primary-camera video and 98.75% accuracy from the wrist-camera video. This check confirms that the re-rendering changes the predicted future in the intended way.

Changing object state while averaging over robot state produces a mean action effect of 0.00015 ([−0.00012, 0.00049]) and a mean executed goal-state effect of 0.00018 ([−0.00040, 0.00102]). Identifying the intended 2×2 target cell from the executed endpoint has an accuracy of 0.25, exactly equal to chance. The experiment therefore finds no detectable action or endpoint effect from the isolated object-state manipulation, even though the intended change can be decoded from the videos. The available analysis does not include the robot-state main effect, so this experiment isolates only the effect of changing object state. In addition, the re-rendered videos may fall outside the distribution of futures normally generated by the policy.

5.3 Natural matched-pair analysis

To avoid synthetic combinations of robot and object state, we also search for futures generated by the unmodified policy that differ primarily in one of these quantities. Only four states meet the matching criteria, fewer than the prespecified minimum of ten. In these states, transplanting a future that differs in robot motion produces a mean action effect of 0.291 ([0.060, 0.521]) and a mean executed robot-endpoint effect of 0.315 ([0.059, 0.571]). For futures that differ primarily in object state, the corresponding action and goal-endpoint means are both 0.021, with confidence intervals that include zero. The exact interval bounds are unavailable, and nearest-donor accuracy remains at chance. We therefore treat this four-state analysis as exploratory.

Across these experiments, action changes are associated most strongly with visible differences in robot motion. Wrist- and primary-camera futures shift action toward the donor, while future proprioception and isolated changes to object state have estimated effects near zero. The four natural matched pairs show the same pattern without re-rendering. We refer to this pattern as prospective visual action conditioning. We have not directly decoded a latent representation of robot motion, and the same content experiment has not been conducted in Cosmos 3.

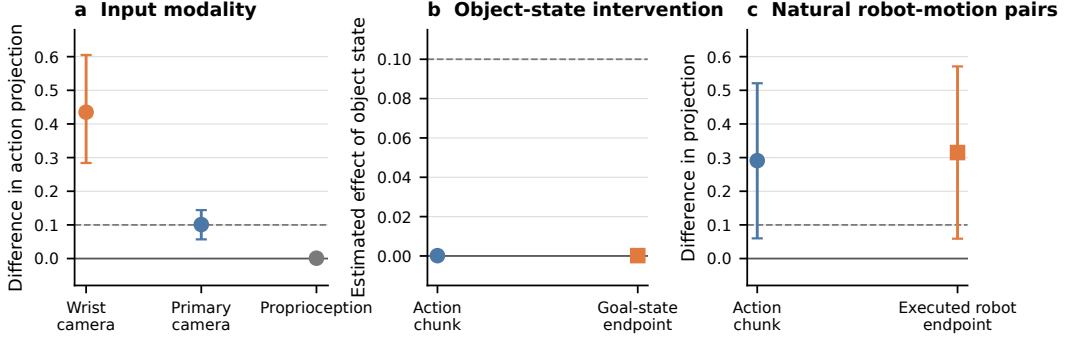


Figure 3: Content interventions in Cosmos Policy. (a) Wrist- and primary-camera future replacement shifts action toward the donor, while future proprioception has an estimated effect near zero. (b) Changing object state has almost no effect on action or executed goal-state projection, even though a decoder reliably recovers the manipulated state. (c) In four exploratory natural pairs, futures that differ in robot motion shift both actions and executed robot endpoints. Bars show 95% confidence intervals obtained by resampling states.

244 6 Analysis of Future-Token Keys and Values in Cosmos 3

245 6.1 Replacing future-token keys and values

246 We next test whether the keys and values computed for future-video tokens contribute to the action
 247 change observed after future transplantation. We first run the model with the recipient future and
 248 record the future-token keys and values. We then run the model with the donor future and replace
 249 selected future-token keys and values with those recorded from the recipient run. No tokens are
 250 added, removed, or reordered. The current-video, text, and action-token keys and values remain
 251 unchanged, as do the current observation, instruction, and noise realization. Replaying a recorded
 252 run and replacing a recipient run’s keys and values with the same keys and values both reproduce
 253 the original action exactly.

254 We evaluate three replacement schemes. The first replaces future-token keys and values only at
 255 layer 16. The second replaces them at every attention connection through which action tokens
 256 read future tokens directly. The third also replaces them where current-video tokens read future
 257 tokens, thereby blocking the indirect route in which future information first enters the current-video
 258 representation and later reaches the action. For each scheme, we report how much the replacement
 259 decreases projection toward the donor action. This intervention-specific decrease is not an estimate
 260 of a population natural indirect effect [16, 22].

261 6.2 Calibration results

262 We first evaluate the three replacement schemes on four calibration tasks that are excluded from the
 263 held-out evaluation. Without key-value replacement, the mean projection toward the donor action
 264 is 0.981. Replacing future-token keys and values at layer 16 decreases projection by 0.346 units
 265 ([0.192, 0.501]). Replacing them at every direct future-to-action connection decreases projection by
 266 0.842 units ([0.766, 0.910]), with a positive decrease in all four tasks. Replacing them along both
 267 the direct route and the indirect route through current-video tokens decreases projection by 0.995
 268 units ([0.976, 1.013]). All record, replay, and same-run replacement controls reproduce the original
 269 actions exactly.

270 6.3 Held-out evaluation

271 After completing the calibration analysis, we select layer 16 and evaluate the same replacement
 272 schemes without modification on the held-out Banana state (Figure 4b). When the donor future is
 273 model-predicted, action projection is 0.900 without replacement. It changes to 0.749 after replace-
 274 ment at layer 16, 0.122 after replacement along all direct future-to-action connections, and -0.012
 275 after also blocking the indirect route through current-video tokens. When the future is encoded from

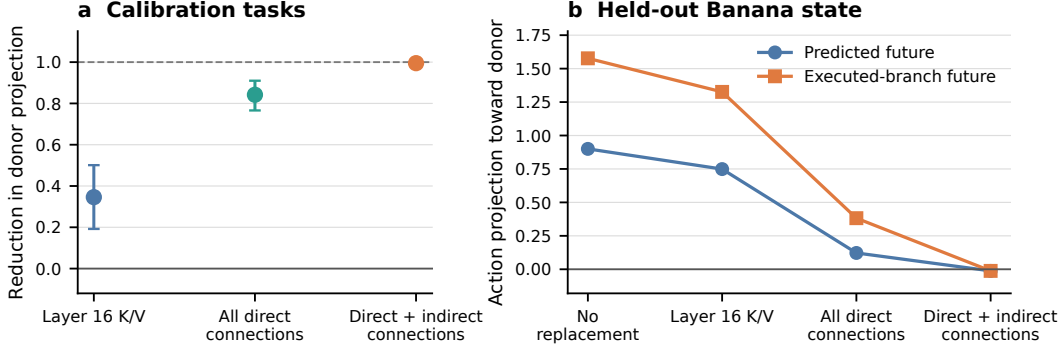


Figure 4: Replacing future-token keys and values in Cosmos 3. (a) Substituting keys and values from the recipient run for the donor run’s future-token keys and values reduces donor projection in all four calibration tasks. Bars show 95% confidence intervals obtained by resampling tasks. (b) The same replacements reduce action projection on the held-out Banana state for both predicted futures and futures encoded from simulator-executed branches.

276 a simulator-executed donor branch, the corresponding action projections are 1.577, 1.326, 0.382,
 277 and -0.012 .

278 The endpoint results are less consistent. For the predicted future, endpoint projection changes from
 279 0.973 without replacement to 0.242 after replacing the direct future-to-action connections and to
 280 -0.335 after also blocking the indirect route. For the future encoded from an executed branch, the
 281 direct replacement changes endpoint projection only from 1.019 to 0.976, despite its larger effect
 282 on the action. Thus, replacing future-token keys and values reduces the action effect for both future
 283 sources in this state, but does not consistently reduce the executed endpoint effect. Repeating the
 284 calibration in five independently launched processes yields bitwise-identical action effects. These
 285 repetitions verify numerical stability but do not provide additional independent states.

286 Replacing future-token keys and values substantially reduces the action change. This shows that
 287 these tensors contribute to the effect, but does not show that they are the only way information from
 288 the future reaches the action [18]. Appendix C reports an additional attention experiment in Cosmos
 289 Policy.

290 7 Discussion

291 In both models, future transplantation shifts action toward the behavior associated with the replace-
 292 ment future. This extends the RIFT result that changing values at future positions affects action
 293 generation [21]. An arbitrary perturbation may change the action in any direction; the positive pro-
 294 jections toward the donor instead show that the content of the replacement future matters.

295 In Cosmos Policy, the size of the effect depends on what changes in the future. The effect is largest
 296 for the wrist camera, which directly shows robot motion, and the estimated effects of future propri-
 297 oception and isolated object-state changes are near zero. Natural pairs that differ in robot motion
 298 produce positive effects without using re-rendered videos. We call this pattern prospective visual
 299 action conditioning. It resembles visual inverse dynamics, in which an action is inferred from a cur-
 300 rent observation and a target future observation [1, 7]. We do not directly decode or locate a latent
 301 representation of robot motion.

302 The content and internal experiments cannot yet be combined into one mechanism because they
 303 were conducted in different models. Cosmos Policy shows which changes to a future affect action.
 304 Cosmos 3 shows that future-token keys and values contribute to its action change. We do not show
 305 that the visible robot-motion information identified in Cosmos Policy is represented in the tensors
 306 replaced in Cosmos 3.

307 These experiments do not test whether either model plans. Visual planning methods generate candi-
 308 date trajectories or subgoals, evaluate their predicted outcomes using a goal or value function, and
 309 choose among them before acting [2, 3, 8]. We do not test candidate search, value comparison, or

310 consistent action selection based on predicted task outcomes. We use the term “imagined future”
311 only to refer to a future representation generated by the model.

312 **Broader impact.** Future transplantation may help determine whether a model’s visual predictions
313 affect its actions, rather than merely appearing plausible. The same intervention could also be used
314 to redirect actions, and effects measured in a small number of simulator states may not persist in
315 deployment. Before using such interventions in deployed systems, they would need to be tested
316 across more states and combined with physical safety constraints and action monitoring.

317 8 Limitations

318 The experiments cover a limited number of independent states. The Cosmos Policy study includes
319 ten LIBERO tasks, but the large effects occur at the first evaluation state in each task. Effects are
320 much smaller after three and six action chunks. Because each evaluation point uses a different
321 saved state, the comparison changes both the state and the amount of elapsed execution. It cannot
322 determine whether the reduction is caused by time, by the particular later states, or by another
323 difference between the groups. The Cosmos 3 donor experiment contains four independent states,
324 and the key–value replacement experiment contains one additional held-out state. These sample
325 sizes do not support conclusions about the full distribution of states encountered by either policy.
326 Testing more donors or repeating an intervention from the same state improves measurement within
327 that state but does not add independent evidence. We also do not measure whether the interventions
328 change binary task success.

329 The Cosmos Policy content analysis has additional qualifications. The factorized videos combine
330 robot and object states independently and may be outside the distribution of futures generated by the
331 model. Although the decoder confirms that the intended object changes are present, the available
332 analysis does not include the numerical main effect of changing robot state. The natural matched-
333 pair analysis avoids re-rendered videos but includes only four independent states, fewer than the
334 prespecified minimum of ten. Exact confidence intervals are also unavailable for its object-only
335 estimates. We have not repeated the robot–object experiment in Cosmos 3. The visible-motion
336 interpretation is therefore limited to the sampled Cosmos Policy states and does not show that either
337 model ignores object consequences in general.

338 The held-out Cosmos 3 result also differs between actions and endpoints. Replacing keys and values
339 at the direct future-to-action connections reduces action projection for both future sources, but does
340 not reduce endpoint projection for the future encoded from an executed branch. The Cosmos Policy
341 attention experiment changes the normalization of the remaining attention weights and lacks a re-
342 ported state count, confidence intervals, and endpoint results. Repeating the code in new processes
343 verifies numerical stability, but not generalization to new states. Appendix B reports exploratory and
344 excluded results, and Appendix A lists missing reproducibility details.

345 9 Conclusion

346 Replacing one generated future with another moves action toward the behavior associated with the
347 replacement in both Cosmos Policy and Cosmos 3. The Cosmos Policy content results are most
348 consistent with prospective visible robot motion. In Cosmos 3, the movement becomes much smaller
349 when future-token keys and values are replaced with those from the original run. Because these
350 experiments use different models, we do not yet follow robot-motion information from a generated
351 future through the network to the action. Future work should test this mechanism within one model,
352 at later decision points, and on task outcomes.

353 References

- 354 [1] Yilun Du, Sherry Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Joshua B. Tenenbaum, Dale Schuurmans,
355 and Pieter Abbeel. Learning universal policies via text-guided video generation. In *Advances in Neural*
356 *Information Processing Systems*, volume 36, 2023. URL [https://proceedings.neurips.cc/](https://proceedings.neurips.cc/paper_files/paper/2023/hash/1d5b9233ad716a43be5c0d3023cb82d0-Abstract.html)
357 [paper_files/paper/2023/hash/1d5b9233ad716a43be5c0d3023cb82d0-Abstract.](https://proceedings.neurips.cc/paper_files/paper/2023/hash/1d5b9233ad716a43be5c0d3023cb82d0-Abstract.html)
358 [html](https://proceedings.neurips.cc/paper_files/paper/2023/hash/1d5b9233ad716a43be5c0d3023cb82d0-Abstract.html).

- [2] Yilun Du, Sherry Yang, Pete Florence, Fei Xia, Ayzaan Wahid, Brian Ichter, Pierre Sermanet, Tianhe Yu, Pieter Abbeel, Joshua B. Tenenbaum, Leslie Kaelbling, Andy Zeng, and Jonathan Tompson. Video language planning. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=9pKtcJcMP3>.
- [3] Frederik Ebert, Chelsea Finn, Sudeep Dasari, Annie Xie, Alex Lee, and Sergey Levine. Visual foresight: Model-based deep reinforcement learning for vision-based robotic control. *arXiv preprint arXiv:1812.00568*, 2018. URL <https://arxiv.org/abs/1812.00568>.
- [4] C. A. Field and A. H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x.
- [5] Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching, 2023. URL <https://arxiv.org/abs/2304.05969>. arXiv preprint, version 2.
- [6] Yanjiang Guo, Yucheng Hu, Jianke Zhang, Yen-Jen Wang, Xiaoyu Chen, Chaochao Lu, and Jianyu Chen. Prediction with action: Visual policy learning via joint denoising process. In *Advances in Neural Information Processing Systems*, volume 37, pages 112386–112410. Curran Associates, Inc., 2024. doi: 10.52202/079017-3570. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/cbe25fa0e7c7084049276888a09acc8d-Abstract-Conference.html.
- [7] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 24328–24346. PMLR, 2025. URL <https://proceedings.mlr.press/v267/hu25g.html>.
- [8] Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang, Shuran Song, Ming-Yu Liu, Chelsea Finn, and Jinwei Gu. Cosmos Policy: Fine-tuning video models for visuomotor control and planning. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=wPEIStHxYH>.
- [9] Daniël Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177.
- [10] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model, 2025. URL <https://arxiv.org/abs/2503.00200>. arXiv preprint, version 3.
- [11] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 44776–44791. Curran Associates, Inc., 2023. doi: 10.52202/075280-1939. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/8c3c666820ea055a77726d66fc7d447f-Abstract-Datasets_and_Benchmarks.html.
- [12] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. doi: 10.52202/068431-1262. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html.
- [13] Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-scale simulation of household tasks for generalist robots. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024. doi: 10.15607/RSS.2024.XX.050. URL <https://www.roboticsproceedings.org/rss20/p050.html>.
- [14] NVIDIA. Cosmos 3: Omnimodal world models for physical AI. *arXiv preprint arXiv:2606.02800*, 2026. doi: 10.48550/arXiv.2606.02800. URL <https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf>.
- [15] NVIDIA. Cosmos3-nano-policy-droid. Hugging Face model checkpoint and configuration, 2026. URL <https://huggingface.co/nvidia/Cosmos3-Nano-Policy-DROID>. Checkpoint revision 2b9f9517.
- [16] Judea Pearl. Direct and indirect effects. In *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence*, pages 411–420, San Francisco, CA, 2001. Morgan Kaufmann. URL https://ftp.cs.ucla.edu/pub/stat_ser/R273.pdf.

- [17] Lu Qiu, Yizhuo Li, Yi Chen, Yuying Ge, Yixiao Ge, and Xihui Liu. Making foresight actionable: Repurposing representation alignment in world action models, 2026. URL <https://arxiv.org/abs/2606.12217>. arXiv preprint.
- [18] Sankaran Vaidyanathan, David Arbour, Aaron Mueller, Scott Niekum, and David Jensen. The curse of multiple mediators: Hidden interaction effects in activation patching, 2026. URL <https://arxiv.org/abs/2606.27510>. arXiv preprint.
- [19] Xuning Yang, Rishit Dagli, Alex Zook, Hugo Hadfield, Ankit Goyal, Stan Birchfield, Fabio Ramos, and Jonathan Tremblay. RoboLab: A high-fidelity simulation benchmark for analysis of task generalist policies. In *Proceedings of Robotics: Science and Systems*, Sydney, Australia, July 2026. doi: 10.15607/RSS.2026.XXII.096. URL <https://roboticsproceedings.org/rss22/p096.html>.
- [20] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination?, 2026. URL <https://arxiv.org/abs/2603.16666>. arXiv preprint, version 2.
- [21] Chushan Zhang, Jinguang Tong, Xuesong Li, Yikai Wang, and Hongdong Li. Keep the future, drop the rollout: RIFT for world action models, 2026. URL <https://arxiv.org/abs/2608.11521>. arXiv preprint, version 2.
- [22] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Hf17y6u9BC>.
- [23] Weiheng Zhao, Haoyi Jiang, Xin Shi, Liu Liu, Fan Huang, Zhizhong Su, Wei Sui, and Xinggang Wang. Faster-WAM: Efficient inference-time future conditioning for robust world action models, 2026. URL <https://arxiv.org/abs/2608.04404>. arXiv preprint.

A Reproducibility Details

The Cosmos Policy experiments use public commit 18a2acca, checkpoint revision cb689ec0, LIBERO, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. Endpoint vectors concatenate goal-relevant simulator coordinates with the official nine-dimensional proprioceptive endpoint. The prespecified design includes ten tasks, three decision points, eight unmodified policy runs per state, one reciprocal donor pair per state, and three future-noise draws per direction. Confidence intervals use 10,000 bootstrap resamples of simulator states.

The Cosmos 3 experiments use checkpoint revision 2b9f9517. The public model report and checkpoint describe a 36-layer Nano policy that uses four diffusion steps and predicts a 32-action chunk of absolute joint positions [14, 15]. For each endpoint evaluation, we reconstruct the simulator, restore the saved state, and replay the same observation prefix. Separate control runs check the token layout, sampler wrapper, cache recording and replay, and same-run replacement. Each control reproduces the expected action bitwise, and sentinel values confirm that interventions to future representations never modify the action-token slice directly.

Complete reproduction additionally requires the full seed list, repository hashes, container digests, per-run wall-clock times, and aggregate compute. These items are not included in the current artifact. The provided `results/paper_summary.json` contains the values used to generate the manuscript figures, but not the full intervention logs or implementation code.

B Additional Results and Exclusions

We separately evaluated six states in RoboCasa [13]. This experiment was exploratory and is not pooled with the LIBERO analysis. The donor-minus-self endpoint effect is 0.0287 (95% CI [0.0101, 0.0492]) and is positive in all six states. The action effect is smaller and does not reliably exceed the Gaussian control. The selected Predict2 continuation pairs do not show stable differences in binary success across repeated runs. An unfinished 20-state Cosmos 3 study is excluded from all reported estimates.

C Future-to-Action Attention in Cosmos Policy

We also remove future-token keys from action-token queries at block 27 of Cosmos Policy. The available experiment summary reports four intervention settings, 0.25, 0.50, 0.75, and 1.00, but does not specify the gating rule used to define them. Across these settings, the normalized action L2 change is 0.022, 0.035, 0.054, and 0.068. Recomputing attention with all keys retained reproduces the original result bitwise. At the 1.00 setting, the action change is 0.016 larger than when the same number of keys is chosen at random and 0.048 larger than when future keys are removed at block 0. Removing the same number of current-observation keys at block 27 produces an effect 0.024 larger than removing future keys.

Removing keys also renormalizes attention over the keys that remain, so this experiment does not isolate the contribution of the removed connections alone. The available results also omit the number of independent states, confidence intervals, and endpoint effects. We therefore treat this experiment only as supporting evidence that future-to-action attention at block 27 affects action generation.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Sections 1, 4–9 state the future-replacement result, its variation across decision points, and the limits on claims about planning and task success.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Result-specific qualifications appear in Sections 4–6, and Section 8 discusses model and state coverage, the later-state comparison, the factorized object intervention, the internal replacement experiments, and the absence of task-success results.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper reports empirical intervention results and does not claim a theorem or formal proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Sections 3–6 and Appendix A specify the estimands, experimental units, controls, clustering, public checkpoints, and exactness gates. The complete seed list, all repository hashes and container digests, intervention code, and run logs are not yet included.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The source package includes machine-readable values for the paper figures, but not the intervention code, raw logs, or an anonymous public reproduction artifact.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sections 3–6 provide the state grid, branch construction, intervention targets, controls, action horizons, model interfaces, and statistical analysis needed to interpret the results. Appendix A states which exact environment details are available and which remain unreported.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Principal estimates have explicitly defined 95% cluster-bootstrap intervals, the independent unit and 10,000-resample procedure are stated in Section 3.4, and exact sign counts are reported where relevant. The text identifies controls and secondary means for which exact intervals were not available.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: Appendix A reports GPU type and software environment, but complete per-run wall-clock and aggregate compute-hour accounting is still being assembled for the submission artifact.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

Answer: [Yes]

Justification: The experiments use public robot-policy checkpoints and simulation benchmarks, involve no people or personal data, and report negative and failed intervention results rather than selectively discarding them.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 7 discusses auditability benefits, action-redirection misuse, false confidence from narrow simulation, and corresponding validation and safety-monitoring needs.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: This work does not release a new high-risk pretrained model or scraped dataset; it analyzes existing robot policies in simulation.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: Existing assets are cited and exact source revisions are recorded, but an explicit license inventory for every upstream code and benchmark asset remains to be added before submission.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [No]

Justification: New intervention code and result manifests are documented in the working artifact, but the anonymized release package and its final license are not yet attached to this draft.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The research contains no crowdsourcing and no human participants.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

791 Answer: [N/A]
 792 Justification: The research contains no human-subject data collection or interaction.
 793 Guidelines:
 794 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
 795 with human subjects.
 796 • Depending on the country in which research is conducted, IRB approval (or equiva-
 797 lent) may be required for any human subjects research. If you obtained IRB approval,
 798 you should clearly state this in the paper.
 799 • We recognize that the procedures for this may vary significantly between institutions
 800 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
 801 guidelines for their institution.
 802 • For initial submissions, do not include any information that would break anonymity
 803 (if applicable), such as the institution conducting the review.
 804 **16. Declaration of LLM usage**
 805 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 806 non-standard component of the core methods in this research? Note that if the LLM is used
 807 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
 808 scientific rigor, or originality of the research, declaration is not required.
 809 Answer: [N/A]
 810 Justification: Language models were used only for writing and formatting assistance, not
 811 as a component of the experimental method or scientific evaluation.
 812 Guidelines:
 813 • The answer [N/A] means that the core method development in this research does not
 814 involve LLMs as any important, original, or non-standard components.
 815 • Please refer to our LLM policy in the NeurIPS handbook for what should or should
 816 not be described.